Calibration of a neural network ocean closure for improved mean state and variability

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m2lines.github.io

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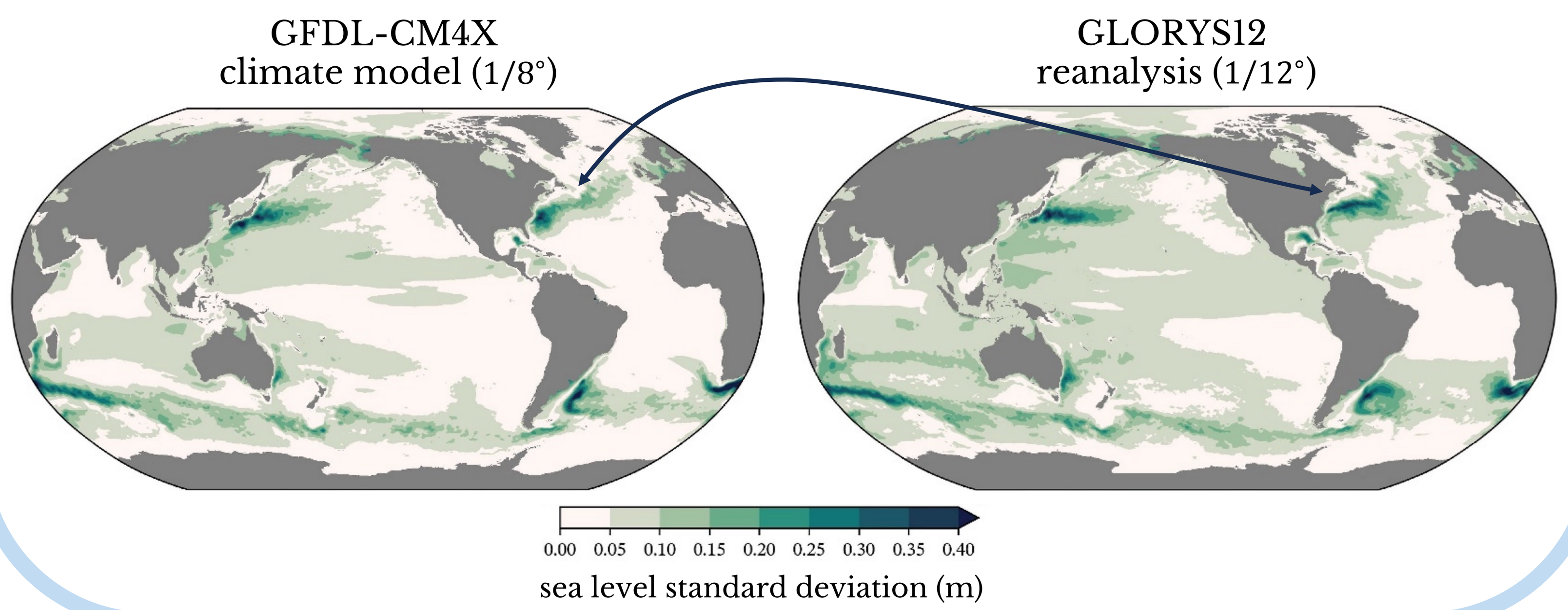
Motivation

Ocean components of climate models demonstrate biases in the **mean state** and **variability** even at relatively high resolutions ($1/4^\circ - 1/8^\circ$)

Griffies, Stephen M., et al. "The GFDL-CM4X climate model hierarchy, Part I: Model description and thermal properties." *Journal of Advances in Modeling Earth Systems* (2025)

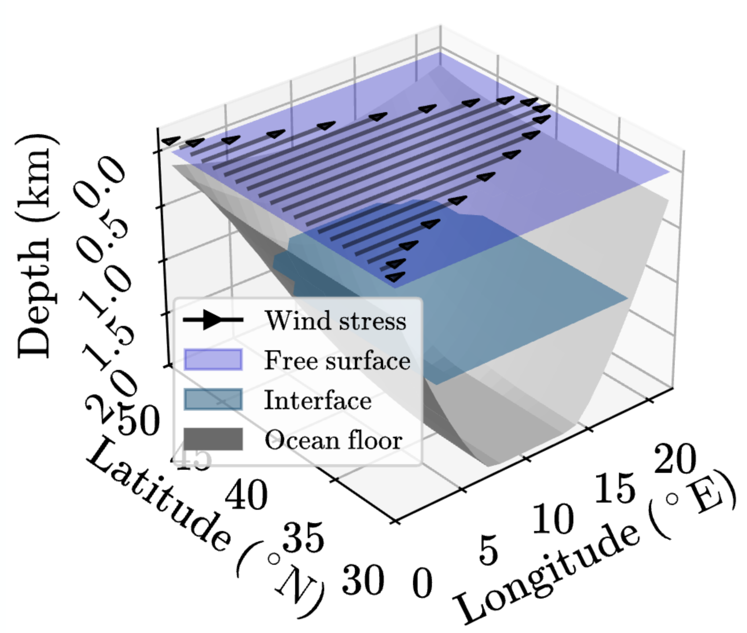


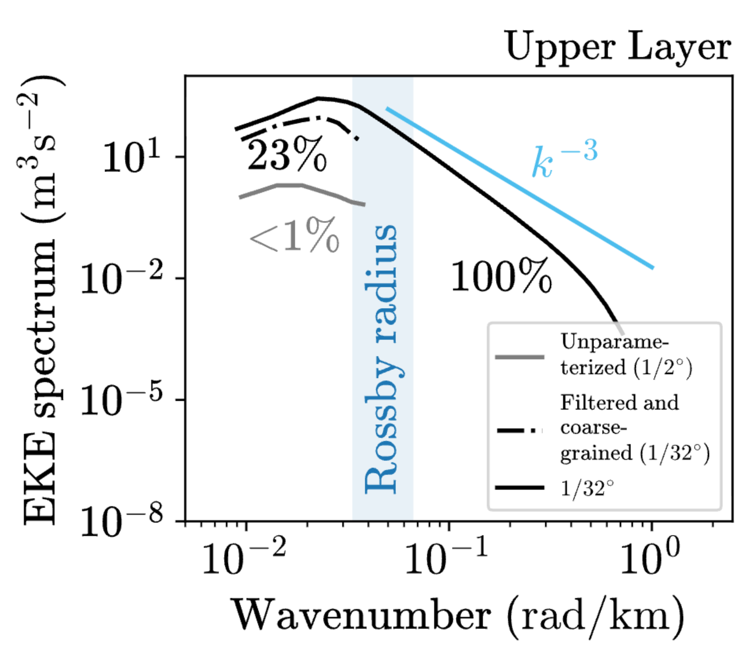
Challenging idealized setup

GFDL MOM6 2-layer wind-driven Double Gyre (DG) simulation:

❖ Less than 1% of eddy kinetic energy (EKE) is reproduced in unparameterized simulation at resolution 1/2°

❖ Only 23% of EKE is resolved in filtered and coarsegrained data





Biases are addressed by invoking neural network (**eANN**) parameterization of **mesoscale eddies** by Perezhogin, Adcroft & Zanna, GRL (2025):

$$\partial_t \mathbf{u} = \cdots + \nabla \cdot \mathbf{T}$$

$$\mathbf{T}(\mathbf{X}, \Delta) = \gamma \Delta^2 ||\mathbf{X}||_2^2 f_\phi(\mathbf{X}/||\mathbf{X}||_2)$$

$$\mathbf{X} = \begin{pmatrix} [\overline{\sigma}_S] \updownarrow 9 \\ [\overline{\sigma}_T] \updownarrow 9 \\ [\overline{\omega}] \updownarrow 9 \end{pmatrix} \in \mathbb{R}^{27}$$

$\overline{\sigma}_S = \partial_y \overline{u} + \partial_x \overline{v}$ – shearing strain,

$\overline{\sigma}_T = \partial_x \overline{u} - \partial_y \overline{v}$ – horizontal tension/stretch,

$\overline{\omega} = \partial_x \overline{v} - \partial_y \overline{u}$ – relative vorticity.

Improving mean state and variability as a calibration problem

We consider optimization problem with loss function:

$$\mathcal{L}(\theta) = ||y - \mathcal{G}(\theta)||_2^2$$

y is the **observational vector**, i.e. statistics of filtered and coarsegrained high-resolution simulation

θ is the **parameter vector**, i.e. weights and biases of the last layer of the neural network

$\mathcal{G}(\theta)$ is the **forward model**, i.e. statistics of coarse parameterized simulation

Statistics are defined as follows:

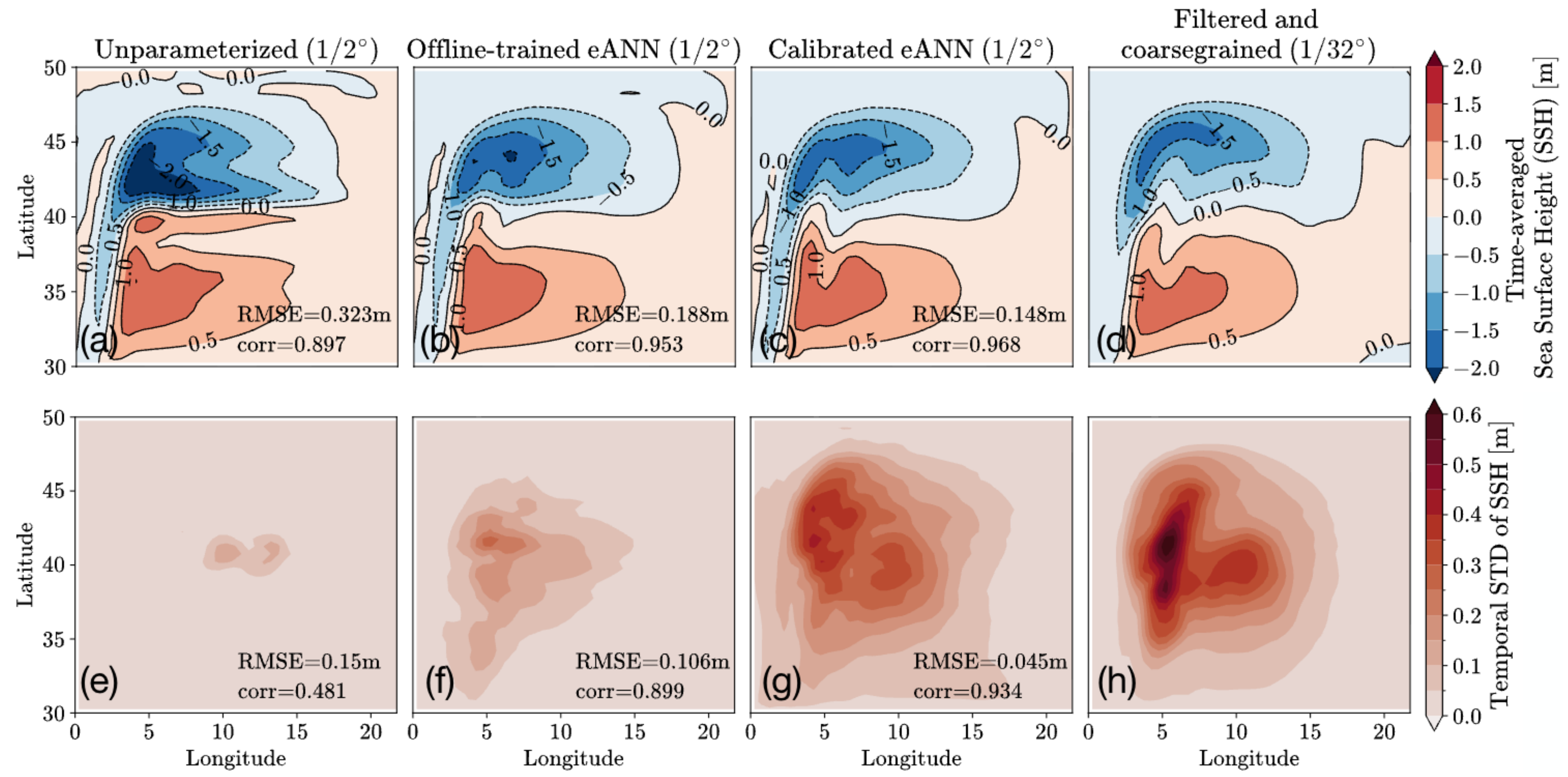
$$y = \begin{bmatrix} \overline{\eta}^t \\ \sqrt{\overline{\eta'^2}^t} \end{bmatrix}$$

$\overline{\eta}^t$ is 10-years averaged fluid interfaces

$\sqrt{\overline{\eta'^2}^t}$ is the standard deviation of fluid interfaces

Gradient-free optimization with Ensemble Kalman Inversion (EKI[†])

Calibrated parameterization reduces errors in mean state and variability by factors of **2.2** and **3.3** relative to unparameterized model



Ensemble-mean update:

$$\overline{\theta}_{j+1} = \overline{\theta}_j + \Theta_j (I + G_j^T G_j)^{-1} G_j^T (y - \overline{g}_j)$$

Ensemble-perturbations update:

$$\Theta_{j+1} = \Theta_j (I + G_j^T G_j)^{-1/2}$$

$\overline{\theta}_j$ is mean parameter vector

$\overline{g}_j$ is mean forward model output

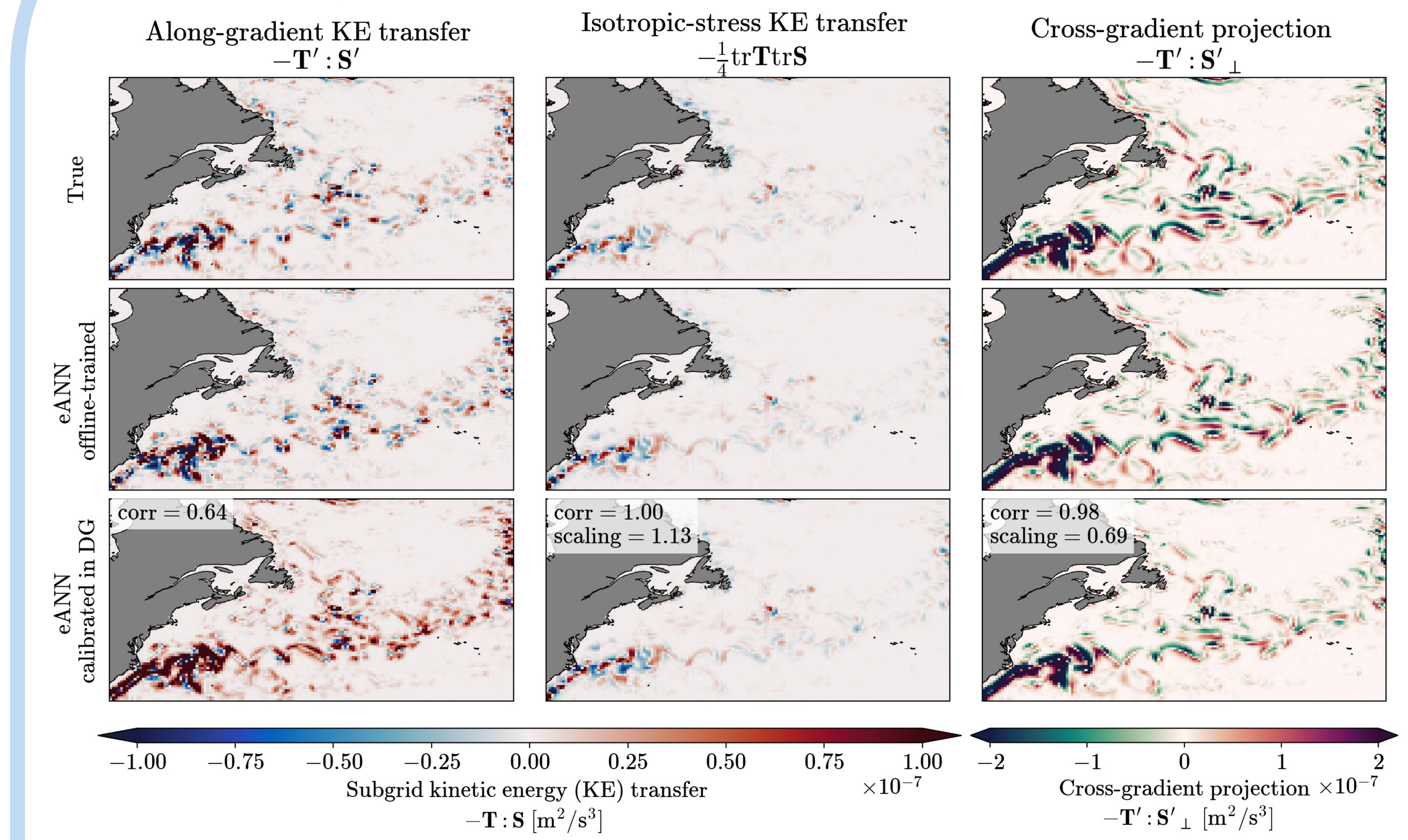
Θ_j, G_j are ensemble perturbations

†Iglesias, Marco A., Kody JH Law, and Andrew M. Stuart. "Ensemble Kalman methods for inverse problems." *Inverse Problems* (2013)

†Huang, Daniel Zhengyu, et al. "Efficient derivative-free Bayesian inference for large-scale inverse problems." *Inverse Problems* (2022)

†Gjini, Rebecca, et al. "The Ensemble Kalman Inversion Race." *arXiv* (2025)

Physical effect of calibration



Calibration **modifies** offline-trained parameterization as follows:

❖ Along-gradient fluxes are **skewed** towards **stronger backscatter** (compensation of numerical viscosity)

❖ Isotropic stress is **unchanged** (as it is less important)

❖ Cross-gradient fluxes are **attenuated** (adjust effective resolution)

Conclusions

- **Mean state and variability can be improved** by calibrating data-driven parameterization of **mesoscale eddies**
- Calibrated parameterization learns **new spatial patterns** which cannot be obtained by simple retuning of the scaling coefficient
- EKI allows to solve optimization problem for **noisy non-differentiable forward models**
- See our paper[#] for scaling to **more challenging setup** (NeverWorld2) and a method to **avoid integration to statistical equilibrium**

Perezhogin, Pavel, Alistair Adcroft, and Laure Zanna. "Calibration of a neural network ocean closure for improved mean state and variability." *arXiv* (2026), under review GRL.

Poster PDF

Paper PDF